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Validity of the Hawkin Dynamics Wireless Dual Force Platform System Against a Piezoelectric Laboratory Grade System for Vertical Countermovement Jump Variables

Thomas Dos'Santos, Daniel T. Evans, and Dale B. Read

Department of Sport and Exercise Sciences, Manchester Institute of Sport, Manchester Metropolitan University, Manchester, United Kingdom

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Abstract

Dos'Santos, T, Evans, DT, and Read, DB. Validity of the Hawkin dynamics wireless dual force platform system against a piezoelectric laboratory grade system for vertical countermovement jump variables. *J Strength Cond Res* 38(6): 1144–1148, 2024—The aim of this study was to determine the criterion validity of the Hawkin Dynamics (HD) wireless dual force platform system for assessing vertical countermovement jump (CMJ) variables, compared with those derived from a Kistler piezoelectric laboratory grade force platform system. During a single testing session, HD force platforms were placed directly on top of 2 adjacent Kistler force platforms to simultaneously collect vertical ground reaction forces produced by 2 male recreational soccer players (age: 29.0 ± 2.8 years, height: 1.79 ± 0.01 m, mass: 85.6 ± 4.7 kg) that performed 25 vertical CMJs each. Sixteen vertical CMJ variables pertaining to jump height (JH), flight time (FT), time-to-take off (TTT), countermovement depth, body weight (BW), propulsive and braking mean, and peak powers, forces, and impulses were compared between systems. Fixed bias was observed for 6 of 16 variables (peak and mean braking power, mean propulsion force, TTT, FT, and BW), while proportional bias was present for 10 of 16 variables (peak and mean propulsive and braking force, TTT, FT, peak and mean braking power, mean propulsive power, and BW). For all variables regardless of fixed or proportional bias, percentage differences were ≤3.4% between force platform systems, with near perfect to perfect correlations (r or ρ = 0.977–1.000) observed for 15 of 16 variables. The HD dual wireless force platform system can be considered a valid alternative to a piezoelectric laboratory grade force platform system for the collection of vertical CMJ variables, particularly outcome (i.e., JH, reactive strength index modified) and strategy variables (countermovement depth).

Key Words: agreement, kinetic, kinematic, force-time

Introduction

The assessment of lower-limb neuromuscular function is an important component of sports science and strength and conditioning practice (11), often used for monitoring acute and longitudinal changes in neuromuscular performance (3). One such method to assess athlete lower-limb impulsive properties is to assess force-time curves (FTC) generated during the vertical countermovement jump (CMJ) performed on a force platform (12). Traditionally, in-ground embedded laboratory force platform systems were used to derive kinetic and kinematic parameters (i.e., jump height [JH], power, impulse) from FTCs to evaluate neuromuscular performance (12); however, they are generally restricted to laboratory environments with the force platforms having to be affixed to the ground.

Importantly, advancements in force platform technology have led to the recent widespread use of cheaper, commercially available, portable, and more accessible force platforms allowing practitioners to collect athlete data quickly in real-world environments. For example, Hawkin Dynamics, Inc (HD), has developed a wireless dual force platform system which has recently been used to monitor changes in neuromuscular function post-soccer match play (14), and notably its hardware has been validated against Advanced Mechanical Technology (AMTI) Inc., laboratory grade force platform (strain gauge) for vertical CMJ, and drop jump force-time measures (1). However, the strain gauge "beam" load cell sensors of the HD force platform system have yet to be validated against a laboratory force platform system using piezoelectric sensors which is a different mechanism of deriving vertical ground reaction force data (vGRF). Thus, the aim of this study was to determine the criterion validity of the HD wireless dual force platform system of selected outcome (e.g., JH, flight time [FT]) and strategy (e.g., time-to-take-off [TTT]) variables during the vertical CMJ, compared with those derived from a piezoelectric laboratory grade force platform system. It was hypothesized that near perfect agreement and minimal differences and bias would be observed between the force platform systems for all CMJ variables.

Methods

Experimental Approach to the Problem

A cross-sectional, within-subject comparative design was employed whereby CMJ variables from a HD wireless dual force

Address correspondence to Thomas Dos'Santos, t.dossantos@mmu.ac.uk.

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platform system were compared against a piezoelectric laboratory grade force platform system to determine criterion validity during a single testing session. Before the vertical CMJs, all subjects performed a 10-minute dynamic warmup. The methodologies employed in this study were based on Badby et al. (1); thus, a brief overview is provided.

Subjects

Two male recreational soccer players (age: 29.0 ± 2.8 years, height: 1.79 ± 0.01 m, mass: 85.6 ± 4.7 kg) who were free from lower limb injury 12 months before testing, and familiar with vertical jump training, partook in this study. Written informed consent was provided by all subjects before participation. Approval for the investigation was provided by the Manchester Metropolitan University Ethics Committee (Ethics approval code: 42015).

Procedures

On arrival, all subjects had their body mass (Seca Digital Scales, Model 707, Birmingham, United Kingdom) and standing height (Seca Stadiometer) measured to the nearest 0.1 kg and 0.01 m, respectively.

Force-Platform Set-Up. A HD wireless dual force platform (HD Inc., fourth generation, ME, USA) was placed directly on top of and within the dimensions of 2 piezoelectric force platforms (Kistler Type 9260AA6, Kistler Instruments, Hampshire, United Kingdom) on a concrete surface (Figure 1) to collect vGRF data independently from each lower limb at 1,000 Hz. Vertical ground reaction force data were collected for 5 seconds using HD proprietary software for the wireless force place system and for 7 seconds using Bioware software (Kistler, Instruments) for the piezoelectric force platform system. Both systems were zeroed before each trial, so the weight of the HD force platform was removed before acquisition. The bilateral force-time data (summed left and right) from both force platform systems for each trial were extracted and analyzed using an identical custom Microsoft Excel spreadsheet designed by the lead researcher, in line with vertical jump analysis recommendations (5,12,13,15). Vertical ground reaction force data from both force platform systems were filtered with a Butterworth low-pass filter (50 Hz cut-off frequency). The braking phase was defined as the period between the instant of peak negative velocity and the instant of zero velocity (1). The propulsive phase was defined as the period between the velocity exceeding 0.01 $m \cdot s^{-1}$ and the instant of take-off (1). Outcome measure definitions are provided in Badby et al. (1). Data were time aligned between force platform systems using touch-down identification, as suggested by Badby et al. (1) due to the Kistler force platform system data acquisition triggered slightly before the HD system. The CMJ variables obtained are presented in Table 1.

Vertical Countermovement Jump. Subjects performed 25 trials of a bilateral vertical CMJ on top of the force platforms, with 1 minute of rest between each trial. Specifically, each subject performed 5 trials each of the following conditions: jump with 50% perceived max effort, jump with 75% perceived max effort, "jump as high as possible," "jump as fast as possible," and "jump as fast and as high as possible" (9) to provide a spectrum of different jump strategies. Subjects were instructed to stand still for an initial 1-second period to permit accurate body weight (BW) calculations for subsequent force-time analysis (i.e., integration) and then were instructed to perform the respective CMJ condition with arms akimbo, keeping limbs extended during flight, and triple flexion of hips, knees, and ankles during the landing.

Statistical Analyses

Mean ± *SD* were calculated for all variables using Microsoft Excel (version 2016, Microsoft Corp., Redmond, WA). As the aim of this study was to compare and assess the agreement of CMJ variables between the 2 force platform systems (HD and Kistler), trials were pooled from each subject and treated as an individual sample to obtain the dependent variables of interest (Table 1). Normality was checked for all dependent variables using a Shapiro-Wilk test using SPSS v28 (SPSS Inc., Chicago, IL). Percentage differences ([Kistler-HD]/average of 2 methods × 100) (6) were calculated between methods using Excel. The mean of the difference between methods was expressed absolutely, and the 95% limits of agreement (LOA) (LOA: mean of the difference ±1.96 *SD*) were calculated using previously described methods (4) in Excel. Pearson's correlations (parametric data) or Spearman's rank correlations (nonparametric data) were also calculated for dependent variables between methods using SPSS and interpreted as trivial (0.00–0.09), small (0.10–0.29), moderate (0.30–0.49), large (0.50–0.69), very large (0.70–0.89), nearly perfect (0.90–0.99), and perfect (1.00) (8). Agreement between methods was determined using Ordinary least products regression (OLPR) (10) with bootstrapped 95% confidence intervals (CI) using SPSS. Intercept 95% CIs that did not include

Figure 1. Experimental set-up for data collection.

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Table 1

Comparison in countermovement jump variables between Hawkin dynamics and Kistler force plates.*†‡

Metric	Kistler force plates		HD force plates		Mean of 2 methods		Mean differences		LOA		% Difference		Correlation
	Mean	*SD*	Mean	*SD*	Mean	*SD*	Mean	*SD*	Lower	Upper	Mean	*SD*	*r* or *ρ*
Pk braking Fz (N)	1888.1	216.2	1933.2	221.2	1910.6	218.7	−45.1	6.4	−57.7	−32.5	−2.4	0.2	1.000**
Pk prop Fz (N)	2041.1	118.7	2088.8	121.8	2064.9	120.2	−47.7	5.5	−58.4	−37.0	−2.3	0.2	*0.998***
Mean braking Fz (N)	1,464.8	150.4	1,499.6	154.1	1,482.2	152.3	−34.8	5.5	−45.7	−24.0	−2.3	0.3	1.000**
Mean prop Fz (N)	1,680.8	109.8	1721.0	114.0	1700.9	111.9	−40.2	6.1	−52.1	−28.3	−2.4	0.3	*0.998***
JH (m)	0.336	0.050	0.341	0.051	0.339	0.050	−0.006	0.005	−0.015	0.004	−1.7	1.4	0.996**
RSImod	0.450	0.075	0.455	0.076	0.452	0.076	−0.005	0.007	−0.018	0.009	−1.0	1.6	0.996**
TTT (s)	0.793	0.105	0.786	0.099	0.789	0.102	0.007	0.013	−0.019	0.033	0.8	1.5	0.993**
FT (s)	0.535	0.046	0.536	0.047	0.536	0.046	−0.001	0.002	−0.005	0.004	−0.1	0.5	*0.998***
CM depth (m)	−0.294	0.052	−0.298	0.053	−0.296	0.052	0.004	0.004	−0.003	0.011	−1.4	1.2	*0.998***
Pk braking power (W)	−1,460.2	308.2	−1,509.5	314.7	−1,484.9	311.4	49.3	11.1	27.6	71.1	−3.4	0.7	1.000**
Pk prop power (W)	4,423.3	379.1	4,563.8	386.3	4,493.5	382.3	−140.5	34.4	−207.8	−73.2	−3.1	0.7	*0.977***
Mean braking power (W)	−1,075.4	214.9	−1,110.2	219.2	−1,092.8	217.0	34.8	6.8	21.6	48.0	−3.2	0.6	*0.999***
Mean prop power (W)	2,347.5	254.3	2,421.9	259.4	2,384.7	256.7	−74.5	17.2	−108.1	−40.8	−3.1	0.7	*0.996***
Braking impulse (N.s)	106.4	15.4	109.7	15.6	108.1	15.5	−3.2	0.8	−4.7	−1.8	−3.0	0.7	*0.997***
Prop impulse (N.s)	216.0	15.3	222.2	15.5	219.1	15.4	−6.2	1.6	−9.3	−3.2	−2.8	0.7	*0.989***
BW (N)	826.3	22.0	842.7	23.8	834.5	22.9	−16.3	2.3	−20.9	−11.8	−2.0	0.2	*0.841***

*BW = bodyweight; CM = countermovement; CI = confidence interval; ES = effect size; FT = flight time; Fz = force; HD = Hawkin dynamics; JH = jump height; LOA = limits of agreement; pk = peak; Prop = propulsion; RSImod = reactive strength index modified; TTT = time-to-take-off.

†Italic denotes nonparametric equivalent statistical test performed.

‡**: $p < 0.001$

0 indicated fixed bias and slope 95% CIs for that did not include 1 indicated proportional bias. The criterion for significance was set at $p \leq 0.05$.

Results

The comparison and agreement in CMJ metrics between force platform systems is presented in Table 1, including means, *SD*, mean difference, LOA, percentage differences, and correlational values. Table 2 presents the OLPR coefficients and bootstrapped 95% CIs. All variables were normally distributed, excluding mean and peak propulsion force, countermovement depth, FT, mean and peak propulsive power output and propulsive and braking impulse, BW, and mean braking power ($p < 0.05$); thus, the nonparametric equivalent statistical test was performed where appropriate. Percentage differences were ≤3.4% between methods for all variables (Table 1), with generally narrow LOAs. Nearly perfect and perfect correlations were observed for 13 and 3 metrics, respectively. Fixed bias was observed for 6 of 16 variables (peak and mean braking power, mean propulsion force, TTT, FT, and BW), while proportional bias was present for 10 of 16 variables (peak and mean propulsive and braking force, TTT, FT, peak and mean braking power, mean propulsive power, and BW) (Table 2). Supplemental Digital Content 1 (see, http://links.lww.com/JSCR/A483) provides Bland-Altman plots for all variables.

Table 2

Ordinary least products regression statistics for countermovement jump metrics between Hawkin dynamics and Kistler force plates.*

Metric	Intercept				Slope			
	Coefficient	95% CI LB	95% CI UB	Fixed bias	Coefficient	95% CI LB	95% CI UB	Proportional bias
Pk braking Fz	−1.604	−12.657	9.449	No	0.977	0.972	0.983	Yes
Pk prop Fz	6.369	−9.455	22.193	No	0.974	0.967	0.982	Yes
Mean braking Fz	0.994	−9.481	11.468	No	0.976	0.969	0.983	Yes
Mean prop Fz	23.320	3.501	43.138	Yes	0.963	0.952	0.974	Yes
JH	−0.004	−0.012	0.004	No	0.995	0.970	1.020	No
RSImod	0.003	−0.013	0.019	No	0.983	0.950	1.016	No
TTT	−0.041	−0.069	−0.013	Yes	1.061	1.024	1.098	Yes
FT	0.011	0.001	0.022	Yes	0.978	0.959	0.997	Yes
CM depth	0.001	−0.005	0.006	No	0.988	0.968	1.007	No
Pk braking power	17.778	7.724	27.832	Yes	0.979	0.970	0.988	Yes
Pk prop power	−55.374	−180.311	69.563	No	0.981	0.952	1.010	No
Mean braking power	13.036	5.836	20.235	Yes	0.980	0.974	0.987	Yes
Mean prop power	−26.724	−58.765	5.317	No	0.980	0.967	0.994	Yes
Braking impulse	−1.676	−3.478	0.126	No	0.986	0.970	1.002	No
Prop impulse	−4.289	−9.892	1.314	No	0.991	0.966	1.017	No
BW	49.875	40.574	59.175	Yes	0.921	0.911	0.932	Yes

*BW = bodyweight; CI = confidence interval; CM = countermovement; FT = flight time; Fz = force; HD = Hawkin dynamics; JH = jump height; LB = lower bound; pk = peak; Prop = propulsion; RSImod = reactive strength index modified; TTT = time-to-take-off; UB = upper bound.

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Discussion

The aim of this study was to determine the criterion validity of the HD wireless dual force platform system utilizing vertical CMJ variables, compared with those derived from a piezoelectric laboratory grade force platform system. The primary findings were that fixed bias and proportional bias was observed for 6 of 16 and 10 of 16 variables, respectively (Table 2). Importantly, however, percentage differences were low (≤3.4%) between methods, with near perfect to perfect correlations (Table 1), thus supporting the hypotheses. As such, the HD dual wireless force platform system can be considered a valid alternative to piezoelectric laboratory grade force platform system for the collection of CMJ variables, particularly vertical CMJ outcome (i.e., JH, reactive strength index modified [RSImod], FT) and strategy variables (TTT, countermovement depth).

In contrast to Badby et al. (1) who observed no fixed or proportional bias (for the same CMJ variables examined in this study) between the HD and AMTI force platforms, in this study, fixed bias and proportional bias were observed for 6 of 16 variables (Table 2) and 10 of 16 variables, respectively (Table 2), between HD and piezoelectric force platforms. This possible discrepancy could be attributed to Badby et al. (1) directly comparing force platform systems which use strain gauge "beam" load cell sensors unlike this study that compared HD force platforms with piezoelectric force platforms which mechanistically derives vGRF differently. This is a significant distinction because each independent type of load cell possesses a different quality in their measurement detection (16). Piezoelectric sensors work by force being applied to a piezoelectric crystal, and then, an electric charge is converted into a voltage signal by means of a charge amplifier. These sensors typically have a simple structure, high resolution, high amplitude over a wide frequency range, and fast response time (2). However, despite this high resolution, the electric charge is typically volatile (2), which could cause additional sensitivity and thus variability in force-time readings used for the identification of specific measures (e.g., in the 1 second of quiet standing during BW calculation), and identifying which "frame" identifies a "peak" measure (e.g., peak propulsive force). Furthermore, piezoelectric sensors are sensitive to temperature changes, making them limited to the laboratory environment. Alternatively, strain gauge sensors do not suffer from this issue, and separate force platform systems have been reported to offer comparably accurate dynamic and static load measurements (1), with typically good within-system reliability in force-time measures (14).

This study also calculated percentage differences between force platforms systems, with low differences observed for all CMJ variables (≤3.4%) (Table 1). Specifically, outcome (i.e., JH, RSImod) and strategy variable (countermovement depth) displayed no fixed or proportional bias, near perfect agreement, with negligible percentage differences (1.0–1.7%) (Tables 1 and 2). In addition, the bias observed in BW could be deemed as negligible due to no meaningful differences being observed in any outcome measures (i.e., JH), which are directly impacted by BW. Crowder et al. (7) also examined the validity of the HD force platform system against AMTI force platforms; however, their results should be interpreted with caution because it was unclear whether the HD force platforms were placed atop of the AMTI. In addition, they used inconsistent sampling rates (AMTI: 1,200 Hz and HD: 1,000 Hz) and also included arm swing during their vertical jump. Nonetheless, this study observed substantially higher associations (Table 1) and narrower LOAs for JH (r = 0.996, LOA = −0.015, 0.004 m vs. r = 0.863, LOA = −0.071, 0.070 m) compared with Crowder et al. (7), thus highlighting the strong agreement and practical utility for the HD force platform system against a laboratory grade system.

It is worth noting that this study only compared vertical CMJ variables between the HD and Kistler piezoelectric force platform system. Future research should establish the agreement of key force-time metrics during high-impact rebound jump tasks and isometric tasks. In addition, the HD force platform system collects uniaxial force. However, the Kistler force platform system collects triaxial force which could account for some discrepancies in kinetic measures. In addition, each force platform system has its own noise characteristics and thus may require its own unique filtering procedures based on residual analysis. However, it should be noted that the HD system automatically applies a Butterworth low-pass filter (50 Hz cut-off frequency) when exporting force-time data, which prohibited residual analysis to be performed on unfiltered data. Thus, we applied identical filtering procedures for Kistler force platform data which may explain some of the subtle discrepancies. Nonetheless, this study observed near perfect agreement and minimal differences in vertical CMJ variables between methods.

Practical Applications

The HD dual wireless force platform system can be considered a valid alternative to piezoelectric laboratory grade force platform system (i.e., Kistler) for collecting vertical CMJ variables, particularly outcome and strategy metrics. As such, this cheaper, portable dual wireless force platform system can be used confidently by practitioners and researchers to evaluate the lower-limb impulsive properties of athletes via the vertical CMJ. The data collected during such tests can then be accurately used to help individualize training, for talent identification purposes, to assess neuromuscular readiness, for injury risk mitigation profiling, and making return to play from injury decisions. However, practitioners are still advised to standardize the force platform systems between testing occasions.

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